

Gated Recurrent Unit (GRU)

What is a GRU?

A Gated Recurrent Unit (GRU) is a type of recurrent neural network (RNN) designed to process sequential data such as text, speech, and time-series data. GRUs use gates to control which information should be remembered and which information can be forgotten.

Main Components

Component	Purpose
Update Gate (z)	Controls how much new information is added and how much previous information is retained.
Reset Gate (r)	Controls how much of the previous hidden state is used when computing new information.
Hidden State (h)	Stores information from previous time steps and carries useful context forward.

Basic Equations

Update gate: $z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$

Reset gate: $r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$

The GRU then uses these gates to calculate a candidate hidden state and combine it with the previous hidden state to produce the new hidden state.

Why Use GRU?

GRUs are generally simpler than LSTMs because they use fewer gates and parameters. This can make them faster to train while still helping the network learn long-term dependencies in sequential data.

Applications

GRUs can be used for language modeling, text classification, speech processing, machine translation, sensor data analysis, and time-series forecasting.